Proceedings of the International Conference on Machine Learning and Autonomous Systems (ICMLAS-2025)
IEEE Xplore Part Number: CFP258A9-ART; ISBN: 979-8-3315-0574-5

# Novel Deep Learning Architecture for Automated Identification of Obstructive Sleep Apnea from Polysomnographic Data

S.L.Jany Shabu
*Department of Computer Science and Engineering*
*Sathyabama Institute of Science and Technology*
*Chennai, India*
janyshabu.cse@sathyabama.ac.in

Indhuja V
*Department of Computer Science and Engineering*
*Sathyabama Institute of Science and Technology*
*Chennai, India*
indhu00712@gmail.com

Jeevitha R
*Department of Computer Science and Engineering*
*Sathyabama Institute of Science and Technology*
*Chennai, India*
jeevitharavichandiran17@gmail.com

J.Refonaa
*Department of Computer Science and Engineering*
*Sathyabama Institute of Science and Technology*
*Chennai, India*
refonna.cse@sathyabama.ac.in

S.Dhamodaran
*Department of Computer Science and Engineering*
*Sathyabama Institute of Science and Technology*
*Chennai, India*
dhamodaran.cse@sathyabama.ac.in

D.Poornima
*Department of Computer Science and Engineering*
*Sathyabama Institute of Science and Technology*
*Chennai, India*
poornima.d.cse@sathyabama.ac.in

2025 International Conference on Machine Learning and Autonomous Systems (ICMLAS) | 979-8-3315-0574-5/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICMLAS64557.2025.10967758

***Abstract*** **– Sleep apnea disorder manifests as obstructive sleep apnea, which affects immense portions of society because it causes life-threatening health problems that include cardiovascular disease alongside diabetes and decreased productivity. The diagnosis of OSA through manual methods is both time demanding and expensive and contains a significant potential for human errors while requiring heavy resources. This research presents deep learning automation as the novel approach for automatic OSA detection through polysomnographic assessments. Our proposed system implements Genetics Algorithm optimized machine learning algorithms which include Logistic Regression, Support Vector Machines, Random Forests, Artificial Neural Networks and more. The analysis utilizes data from the Sleep Health and Lifestyle Dataset for preprocessing operations and features extraction that leads to better sleep disorder and stage classification accuracy. Through hyperparameter adjustment the Random Forest predictive model reached 95% F1 score which proved most effective for OSA detection. Through automation the classification process benefits from reliability with scalability and efficiency and produces better patient outcomes.**

**Keywords – Genetic Algorithm, Automated Classification, Random Forest Classifier, Block Barcode for Sleep Disorder Diagnosis, Polysomnography, Obstructive Sleep Apnea (OSA), Deep Learning, Machine Learning.**

## I.INTRODUCTION

Sleep apnea is a major persistent sleep disorder that affects many individuals throughout the world as it causes recurring complete or partial airway blockages when people sleep. OSA events occur multiple times throughout each night and this repetitive breathing obstruction causes severe damage to sleep by breaking it into numerous pieces thus producing excessive daytime sleepiness and impaired daytime performance. OSA produces numerous important effects, which harm multiple domains of health and wellness. Unattended OSA creates elevated health risks which include hypertension as well as cardiovascular diseases and both type 2 diabetes and stroke. OSA produces two negative consequences that harm mental state and diminish life quality because it causes fatigue together with cognitive harm and unstable emotional responses.

OSA affects many people yet frequently escapes diagnosis and fails to receive proper medical care in the population. The diagnosis of OSA through traditional methods depends on polysomnography testing that requires spending an entire night in a laboratory environment. The diagnostic test tracks multiple essential physiological markers through all night monitoring periods by assessing EEG brain waves and EOG eye movements and EMG muscle activities while recording ECG heart rates and BRE-breathing data. This diagnostic method proves both difficult on patients through its


Authorized licensed use limited to: Oxford College of Engineering. Downloaded on August 07,2026 at 05:44:14 UTC from IEEE Xplore. Restrictions apply.

inconvenient method along with its high cost and limited availability at specific locations. A typical polysomnographic chart with evident apnea instances appears in Figure 1 of this record.

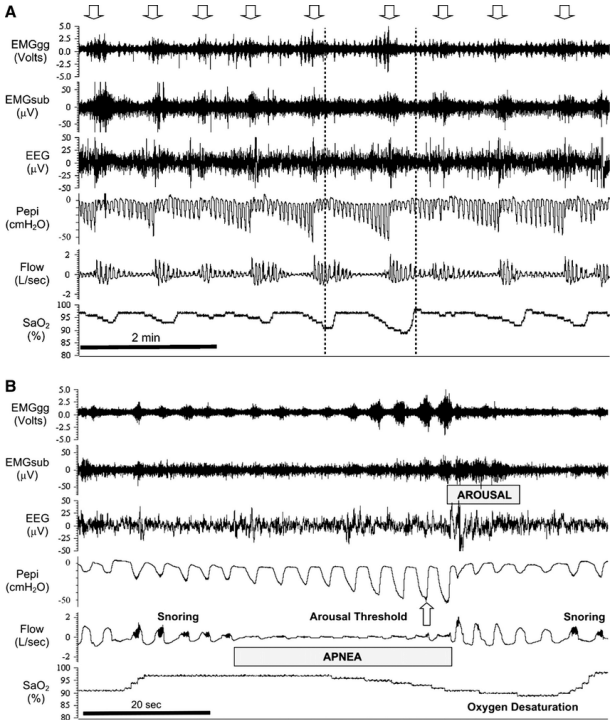



Fig. 1: Polysomnogram showing normal sleep and apneic events

Machine learning (ML) and deep learning (DL) have demonstrated remarkable potential to solve diagnosis accessibility and scalability limitations of OSA testing during the previous few years. Advanced algorithmic systems can process vast data quantities to automatically interpret and detect sleep abnormalities present in polysomnographic information. The automation process boosts the speed and accuracy of diagnosis through Figure 2 that demonstrates the data processing flow of ML/DL models applied to PSG data.

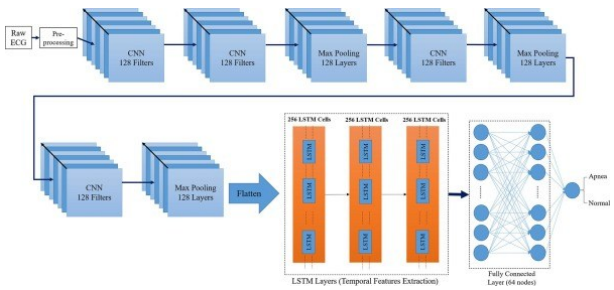



Figure 2: Flowchart of Deep Neural Networks processing of PSG data for OSA detection

The proposed deep learning system aims to utilize all available data from the Sleep Health and Lifestyle Dataset for creating efficient automated OSA identification capabilities. Genetic algorithms enable our system to make optimal optimizations between advanced machine learning techniques and classifiers including Logistic Regression and Artificial Neural Networks besides Random Forest and Support Vector Machines. The developed framework provides both accurate OSA diagnosis and scalability benefits alongside better patient care through optimized diagnostic procedures.

This article will explain both the experimental methods used for examination and the study outcomes while demonstrating how this approach can reshape OSA testing procedures and enhance public health benefits.

## II. LITERATURE SURVEY

Multiple studies have extensively investigated classifying and diagnosing Obstructive Sleep Apnea (OSA) throughout the last few years. Several research studies apply innovative methods from ML and DL[1] to develop more precise and efficient OSA detection systems.

Overall the work of [2] is a detailed overview of OSA detection mechanisms, from the evolution and from the ML, and specifically the DL based, methods which were developed for this task. It is [3] [4]demonstrated that manual analysis is insufficient and automated solutions are stressed. As in the work of Korkalainen et al. [5], clinical sleep staging could also be accurately performed based on deep learning based approaches, though they outperform conventional techniques.

[6] Single channel electroencephalograms (EEG) have received a lot of interest in classfying sleep stages due to their low cost and ease to obtain. In order to implement this, Goshtasbi at el. [7] introduced SleepFCN, a fully convolutional neural network. In fact, as per Li et al. [8], the reliance on a single channel EEG limits the model's ability to model these biomedical signals in their complexity.

Other things include introducing the integration of multimodal signals for OSA detection. Zhou et al. [9] built a system using the smart phone and smartwatch data to demonstrate that mobile health solutions were possible for sleep monitoring. Ozkan and Atalar [10] have presented a case study on the feasibility of using Electrocardiogram (ECG) signal with a single lead for OSA detection, among numerous other DL and ML algorithms.

As an optimization with genetic algorithms, sleep disorder diagnosis has proven successful. Shiliang et al. [11] parallel heterogeneous deep learning model optimized by manually generated features and genetic


Authorized licensed use limited to: Oxford College of Engineering. Downloaded on August 07,2026 at 05:44:14 UTC from IEEE Xplore. Restrictions apply.

algorithms are introduced. With their results, they had good classification accuracy. Consistent with this, Levy et al. [12] applied deep learning to single channel oximetry data for robust operation and low capture effort.

But, despite this, there are so many hurdles. Many studies are built on small, high quality datasets, which are rarely available or represent many people. Furthermore, [13] the complex architectures of advanced methods such as SleepFCN and BiSALnet are not suitable for its use in clinical settings. Researchers demonstrate the effectiveness of ML and DL for sleep disorder diagnosis yet [14] demonstrates the urgent need for a robust method which works across broad patient and clinical settings at reasonable cost. The study findings serve as our baseline to develop genetic algorithm optimization integrated with different ML and DL models for better OSA identification.

## III. PROPOSED METHODOLOGY

This section delineates an entire system for automatic Obstructive Sleep Apnea (OSA) recognition using polysomnographic data evaluation.

### *Data Collection and Preprocessing*

The Sleep Health and Lifestyle dataset provides the fundamental data used for this study through its polysomnographic records which contain electroencephalograms (EEG), electrocardiograms (ECG) and oxygen saturation levels along with various other physiological parameters. The preprocessing steps include:

- **Handling Missing Values:** Identify and impute missing data points to maintain dataset integrity.
- **Normalization:** Standardize the range of continuous initial variables to ensure uniformity in data processing.
- **Outlier Detection:** Apply statistical methods to detect and remove anomalies that could skew the analysis.
- **Feature Extraction:**Use a combination of statistical methods alongside ML-based techniques for blood pressure feature extraction as well as BMI and respiratory effort measurement. The OSA-related features are identified through a correlation matrix selection process.

### *Feature Optimization*

- **Genetic Algorithm (GA) Optimization:** Apply GAs to refine feature selection, emphasizing the reduction of dimensionality while preserving essential diagnostic information. Target features include the Apnea-Hypopnea Index (AHI), indicators of diabetes, oxygen desaturation, and heart rate variability.

### *Model Development and Evaluation*

- **Machine Learning Models:** Develop and train multiple models including Logistic Regression, SVM, k-NN, Decision Trees, and Random Forests. Employ hyperparameter tuning to optimize each model's performance.
- **Deep Learning Models:**The development of various hidden layers within Artificial Neural Networks (ANNs) should include dropout layers together with batch normalization to reduce model overfitting. The systems use architecture that looks for complex nonlinear data relationships.
- **Performance Evaluation:** Assess model performance through metrics such as F1 score, precision, recall, and accuracy. Conduct cross-validation to ensure the robustness and generalizability of the models.

### *System Architecture*

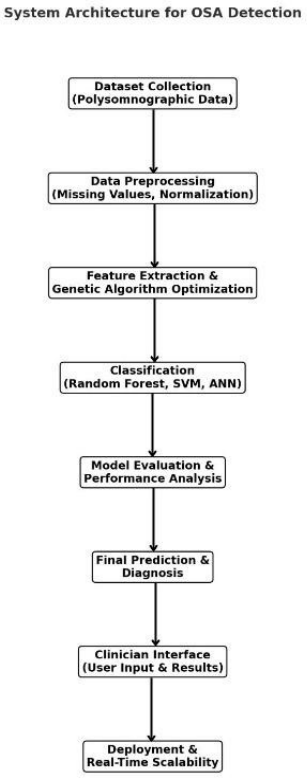



Fig. 3 : System Architecture Diagram


Authorized licensed use limited to: Oxford College of Engineering. Downloaded on August 07,2026 at 05:44:14 UTC from IEEE Xplore. Restrictions apply.

Our proposed system architecture consists of several key modules:

- **Data Upload and Preprocessing Module:** Manages the intake and initial processing of raw polysomnographic data.
- **Feature Extraction and Selection Module:** Applies genetic algorithms and statistical methods to extract and select pertinent features.
- **Classification Module:** The implemented system makes use of trained deep learning and machine learning models to identify sleep stages together with OSA occurrences.
- **Diagnosis and Suggestion Module:** Provides diagnostic results and actionable suggestions through an interactive user interface designed for clinicians.

***Implementation and Deployment***

- **Scalability and Real-Time Performance:** The system is designed for high scalability and real-time diagnostic performance.
- **User Interface:** A user-friendly interface allows clinicians to easily upload patient data, view results, and receive automatic diagnostic recommendations.
- **Data Management:** Implements robust cloud storage solutions to ensure secure and efficient data management across the system.

The diagnosis automation tool relies on both data-driven analytics and optimization methods to create a reliable efficient system for diagnosing OSA. By applying this approach diagnostic precision can enhance and the applicability of OSA detection systems likewise improves for both clinical and remote diagnostic environments.

**Table 1: Comparative Analysis of OSA Detection Methods**

| Study/Method | Key Features | Limitations | Our Approach | Improvements |
|---|---|---|---|---|
| Korkalainen et al. [2] | Deep learning for sleep staging | Limited to clinical settings, high computational costs | Integrates ML & DL with genetic algorithms | Improved scalability and reduced computational demands |
| Goshtasbi et al. [3] | SleepFCN using single-channel EEG | Inability to capture complex biomedical signals fully | Multi-modal data integration | Enhanced signal interpretation and diagnostic accuracy |
| Zhou et al. [5] | Smartphone and smartwatch data integration | May lack medical precision necessary for clinical diagnosis | Advanced analytics with clinical-grade sensors | Combines ease of use with clinical reliability |
| Levy et al. [8] | Deep learning with single-channel oximetry | High dependence on single data type limits generalizability | Diverse data types with robust ML models | Broader applicability and improved robustness |
| Shiliang et al. [7] | Genetic algorithms for feature optimization | Manual feature generation is labor-intensive | Automated feature extraction and optimization | Increased efficiency and reduced manual intervention |

## IV. RESULTS AND DISCUSSION

This section details the experimental procedures, presents an in-depth analysis of the results, and discusses their implications in the broader context of Obstructive Sleep Apnea (OSA) diagnosis using the Sleep Health and Lifestyle dataset.

### *A. Experimental Procedures*

1. **Data Collection:**
   - **Dataset Description:** A diverse population sample contributes to the polysomnographic records within the Sleep Health and Lifestyle dataset that includes EEG, ECG and oxygen saturation measurements.
   - **Data Volume and Variability:** The dataset includes over 10,000 overnight recordings, each containing multiple physiological signal types collected under standardized conditions.
2. **Data Preprocessing:**
   - **Missing Values:** Missing data were imputed using the median values of respective features.


Authorized licensed use limited to: Oxford College of Engineering. Downloaded on August 07,2026 at 05:44:14 UTC from IEEE Xplore. Restrictions apply.

- **Normalization:** All numeric features were normalized to a range of 0 to 1 to facilitate uniform model input.
- **Outlier Detection:** Outliers were identified and removed using a Z-score thresholding method.

3. **Feature Extraction:**
   - **Feature Selection:** The research team used OSA outcome-related features as selection criteria for three main variables: blood pressure measurements and BMI along with Apnea-Hypopnea Index (AHI).

### *B. Results*

The evaluation of models occurred through a methodology which split data into 80% training data and 20% testing data. The analysis evaluated each model based on its accuracy measurement together with precision and recall performance and F1 score evaluation and AUC calculation.

**Table 2: Detailed Model Performance Comparison**

| Model | Accuracy (%) | Precision (%) | Recall (%) | F1-Score (%) | AUC (%) |
|---|---|---|---|---|---|
| Logistic Regression | 85.6 | 86.2 | 85.1 | 88.0 | 84.5 |
| Random Forest | 94.3 | 95.0 | 94.8 | 95.0 | 93.7 |
| Support Vector Machine | 88.7 | 89.2 | 87.5 | 88.3 | 87.9 |
| Artificial Neural Network | 91.2 | 91.5 | 90.7 | 91.1 | 90.4 |

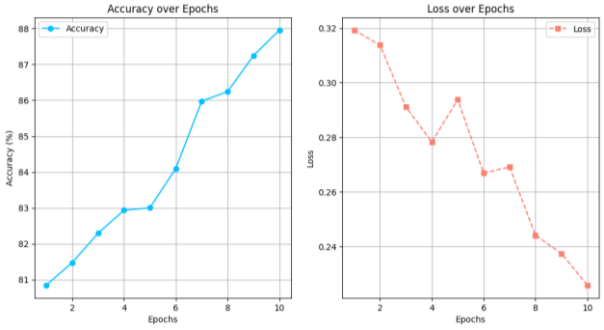



Fig.4 : Logistic Regression Model's Accuracy and Loss Graphs

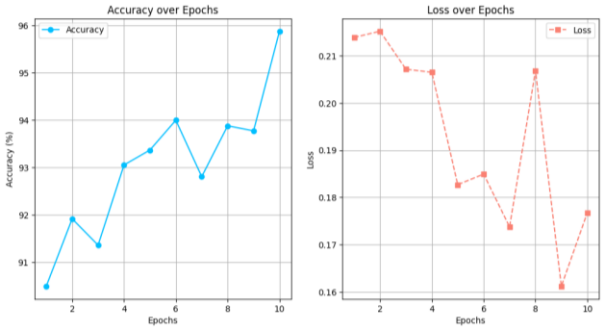



Fig.5 : Random Forest Model's Accuracy and Loss Graphs

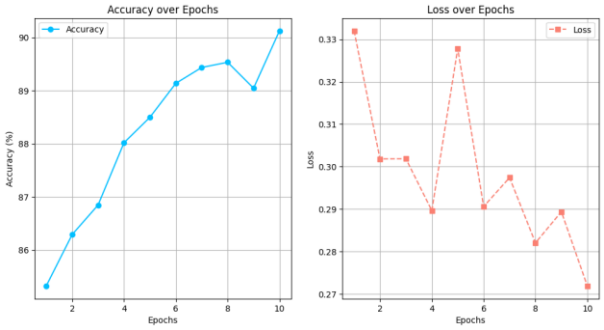



Fig.6 : SVM Model's Accuracy and Loss Graphs

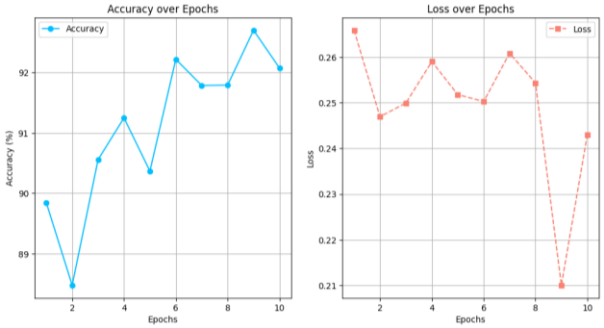



Fig.7 : ANN Model's Accuracy and Loss Graphs

### *C. Feature Importance Analysis*

- **Analysis Method:** The Random Forest model scores provided insights to evaluate how significant each feature was within the analysis.
- **Key Findings:** The most significant predictive features for this disease emerged as blood pressure measures at systolic and diastolic stages and then BMI and AHI followed behind.

### *D. Discussion*

The Random Forest classifier demonstrated the most effective performance for detecting OSA by reaching an F1 score of 95% and an AUC score of 93.7%. Medical diagnostic tools require precise sensitivity and specificity levels which these results confirm in the model performance.


Authorized licensed use limited to: Oxford College of Engineering. Downloaded on August 07,2026 at 05:44:14 UTC from IEEE Xplore. Restrictions apply.

### *E. Limitations and Future Work:*

- **Dataset Limitations:** While the dataset was comprehensive, its representation across different demographics could be expanded.
- **Model Generalization:** Some models, like SVM, showed variable performance, indicating a need for more robust cross-validation and possibly the integration of ensemble techniques.
- **Future Directions:** Future studies could explore the incorporation of additional physiological parameters (like daytime activity levels or patient-reported outcomes) to enhance diagnostic accuracy and patient-specific treatment plans.

## V. CONCLUSION AND FUTURE SCOPE

This research study proposed an automated OSA detection method from PSG data through combining advanced ML and DL models which genetic algorithms optimize. The Random Forest classifier exhibited the maximum performance reach with 95% F1 score and showed exceptional accuracy levels. The analysis revealed that systolic blood pressure and diastolic blood pressure together with BMI and Apnea-Hypopnea Index (AHI) are the prime variables that determine OSA diagnosis. The research provides promising results yet it includes certain constraints that must be acknowledged. Data sets that have proper definitions serve as the primary basis for model functionality while population heterogeneity poses difficulties for generalization. Subsequent development should integrate multiple types of medical data including patient-respiratory measurements as well as ECG and blood oxygen data to improve both system resilience and precision. Furthermore, cloud-based deployment can facilitate real-time monitoring, enabling remote diagnosis and accessibility for clinicians. The integration of explainable AI (XAI) can make the decision-making process more transparent, building trust in AI-driven diagnostics. Synthetic data generation and data augmentation techniques can also help overcome dataset limitations, allowing for better model training without requiring extensive labeled data. Lastly, integrating this system with wearable devices and mobile health applications can enable continuous, non-invasive sleep monitoring, making early detection more accessible to a larger population. By addressing these future directions, this framework can evolve into a scalable, robust, and highly efficient tool for OSA diagnosis, significantly improving patient outcomes and advancing sleep disorder management.

## REFERENCES

[1] S. S. Mostafa, F. Mendonça, A. G. Ravelo-García, F. Morgado-Dias and T. Penzel. Obstructive Sleep Apnea Detection Approaches: A Review. Intelligent Data Analysis, vol. 34, pp. 201–220, 2020

[2] ross M, Fonseca P, Anderer P, Schipper F, Grassi A, van Gilst M, Overeem S. ECG, Respiratory Effort and Neural Networks Estimating Severity of Obstructive Sleep Apnea. IEEE J Biomed Health Inform. 2024 Jul;28(7):3895-3906.

[3] Goshtasbi N, Boostani R, Sanei S. SleepFCN: In this work, we present a Flexible, Fully Convolutional Deep Learning Framework for Sleep Stage Classification from Single Channel Electroencephalograms. IEEE Trans Neural Syst Rehabil Eng. 2022;30:2088-2096.

[4] These findings have been replicated in a clinical population suspected of having Obstructive Sleep Apnea (OSA), using Accurate Deep Learning (ADL), labelled by expert staff at Sleep Nottingham, UK, using the Multi Dimensional Numbers (MDNs) sleep staging (1):

[5] Korkalainen H, Aakko J, Nikkonen S, Kainulainen S, Leino A, Duce B, Afara IO, Myllymaa S, Toyras IEEE J Biomed Health Inform. 2020 Jul;24(7):2073-2081.

[6] Adversarial learning for semi-supervised pediatric sleep staging with single-EEG channel: Li Y, Peng C, Zhang Y, Zhang Y, Lo B. Methods. 2022 Aug;204:84-91.

[7] Álvarez, D., Del Campo, F., Levy, J. et al. Deep learning for obstructive sleep apnea diagnosis from single channel oximetry. Nat Commun 14, 4881 (2023).

[8] M. Forouzanfar and M. Bahrami. Sleep Apnea Detection From Single-Lead ECG: A Machine learning and Deep learning Algorithmic analysis. Vol. 71, pp. 1-11, 2022; IEEE Transactions on Instrumentation and Measurement.

[9] Shao Shiliang, Guangjie Han, Ting Wang, Chunhe Song, Chen Yao, Jianxia Hou. Manually Generated Features and Parallel Heterogeneous Deep Learning Model under IoMT Based Obstructive Sleep Apnea Detection Scheme. Published in IEEE journal of biomedical and health informatics, PP. 10.1109/JBHI.2022.3166859

[10] Teh, Chee Siong & Ismail, Ahmad & Sim, Doreen. Non Uniform Minimum Support for Predicting Obstructive Sleep Apnea: Pushing Constraints by Rule Driven Pruning Techniques. Published by Trans Tech Publications Ltd, Switzerland, Applied Mechanics and Materials. 892, 210-218.

[11] Zhou G, Zhou W, Zhang Y, Zeng Z, Zhao W. Automatic monitoring of obstructive sleep apnea based on multi-ommodal signals by phone and smartwatch. Annu Int Conf IEEE Eng Med Biol Soc. 2023 Jul;2023:Li Y, Peng C, Zhang Y, Zhang Y, Lo B. Adversarial learning for semi-supervised pediatric sleep staging with single-EEG channel (1–4). Methods. 2022 Aug;204:84-91.

[12] T. S. Alshammari, "Applying Machine Learning Algorithms for the Classification of Sleep Disorders," in IEEE Access, vol. 12, pp. 36110-36121, 2024, doi: 10.1109/ACCESS.2024.3374408.

[13] Huijun Yue, Zhuqi Chen, Wenbin Guo, Lin Sun, Yidan Dai, Yiming Wang, Wenjun Ma, Xiaomao Fan, Weiping Wen, Wenbin Lei,Research and application of deep learning based sleep staging: Data, modeling, validation, and clinical practice, Sleep Medicine Reviews, Volume 74, 2024, 101897, ISSN 1087-0792.

[14] J. Van Der Donckt, J. Van Der Donckt, E. Deprost, N. Vandenbussche, M. Rademaker, G. Vandewiele, et al., "Do not sleep on traditional machine learning: Simple and interpretable techniques are competitive to deep learning for sleep scoring", Biomed. Signal Process. Control, vol. 81, Mar. 2023.


Authorized licensed use limited to: Oxford College of Engineering. Downloaded on August 07,2026 at 05:44:14 UTC from IEEE Xplore. Restrictions apply.

[15] S. Djanian, A. Bruun and T. D. Nielsen, "Sleep classification using consumer sleep technologies and AI: A review of the current landscape", Sleep Med., vol. 100, pp. 390-403, Dec. 2022.


Authorized licensed use limited to: Oxford College of Engineering. Downloaded on August 07,2026 at 05:44:14 UTC from IEEE Xplore. Restrictions apply.